

Will AI Replace Bioinformaticians — or Make Them More Powerful?

Introduction

The question is everywhere now; in research laboratories, at conferences, in quiet conversations between graduate students staring at pipelines they half-built with AI-generated code. Artificial intelligence is already changing bioinformatics; that part isn't up for debate. The real question is what that change means for the scientists at the center of it. Does the rise of intelligent machines signal the end of their relevance? Or is it the beginning of something far more interesting?

Bioinformatics sits at the intersection of biology, statistics, and computer science. It was built to make sense of living systems, which are inherently complex, noisy, and vast. For years, that work demanded painstaking human effort, aligning sequences (piecing together DNA fragments like a massive jigsaw puzzle), annotating genomes (labeling what those genes actually do), and interpreting patterns buried inside enormous datasets.

How AI Is Transforming Bioinformatics

That work is shifting. Fast.

AI-powered tools now streamline these repetitive tasks, processing datasets far faster than traditional approaches. AlphaFold, DeepMind's landmark model, solved in months what structural biologists had spent decades working toward (Jumper *et al.*, 2021). The data challenge driving this urgency is staggering, the widespread adoption of high-throughput sequencing technologies has led to an exponential accumulation of genomic, transcriptomic, and proteomic datasets, creating data management and interpretation challenges that continue to intensify (Jiang *et al.*, 2025). Manual analysis simply cannot handle that volume. At that scale, AI isn't just helpful, it is essential.

Beyond speed, AI is expanding what is biologically possible. Foundation models trained on DNA sequences, like the Nucleotide Transformer, can now be fine-tuned to predict regulatory elements and disease-associated variants across the genome (Dalla-Torre *et al.*, 2025). These are not incremental improvements. They represent a fundamental shift in how biological discovery happens.

What Machines Still Cannot Do

But essential tools still need skilled hands.

Bioinformaticians bridge disciplines. They understand both the biological context of the data and the real limitations of the computational tools. That dual fluency is something no model currently replicates. An algorithm can flag a genetic variant as potentially disease-causing. Yet it takes a trained scientist to ask whether that flag is biologically meaningful or statistically misleading. Deep learning models remain opaque, and someone has to be accountable for what the model misses (Zou *et al.*, 2019).

The field is not simply automating old tasks either, it is generating entirely new ones. Interpreting the outputs of multi-omics integration, validating AI-predicted protein interactions, and designing experiments to test computationally derived hypotheses all require human expertise that cannot yet be offloaded.

Conclusion

Replacement was never really the right frame. The field isn't shrinking, it is demanding more; more biological understanding, more computational confidence. The bioinformatician of the future isn't competing with machines. They are directing them, questioning them, and doing the interpretive work no algorithm is equipped to do. That is not a diminished role. It is a profoundly more powerful one. And in genomics, we are only beginning to discover what that looks like.

References

- Dalla-Torre, H., Gonzalez, L., Mendoza-Revilla, J., Lopez Carranza, N., Grzywaczewski, A. H., Oteri, F., and Pierrot, T. (2025). Nucleotide transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, **22**(2), 287-297.
- Jiang, J., Li, Y., Cao, S., Shan, Y., Liu, Y., Fei, T., and Yuan, J. (2025). Artificial intelligence in bioinformatics: a survey. *Briefings in Bioinformatics*, **26**(6), bbaf576.
- Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., and Hassabis, D. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature*, **596**(7873), 583-589.
- Zou, J., Huss, M., Abid, A., Mohammadi, P., Torkamani, A., and Telenti, A. (2019). A primer on deep learning in genomics. *Nature genetics*, **51**(1), 12-18.